$\theta = 1.96°$

2(a) For *vertical* motion and taking *downwards* as positive,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow 20 = 0 + \frac{1}{2}(9.81)t^2$$

$$\Rightarrow t = 2.02\,\text{s}$$

(b) The drone and the package have the same horizontal speed, so they travel the same horizontal distance in 2.02 s. Therefore, the horizontal distance between package and drone is 0 m.

(c) For *vertical* motion and taking *downwards* as positive,
Vertical velocity just before hitting the ground,

$$v_y = u_y + a_y t = 0 + (9.81)(2.02) = 19.8\,\text{m s}^{-1}$$

For *horizontal* motion,
Horizontal velocity just before hitting the ground,

$$v_x = u_x + a_x t = 8.0 + (0)(2.02) = 8.0\,\text{m s}^{-1}$$

Therefore velocity just before hitting the ground,

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{8.0^2 + 19.8^2} = 21.4\,\text{m s}^{-1}$$

Direction of velocity, $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{19.8}{8.0}\right) = 68°\ \text{wrt to horizontal}$

3(a)(i) The statement is not correct. In order for object to come to rest, it has to experience